

ADHD from Brain Waves

Telling ADHD vs. Not From EEG — Done the Honest Way

Capstone Project Report

An AI that reads kids' brain-wave recordings, tested carefully so the 92% score is actually real.

RL · RLcapstone.ai

1. Summary

This project reads EEG recordings (brain-wave signals from sensors on the scalp) from kids and guesses whether each recording came from the ADHD group or the non-ADHD group of a study. I used an AI model that reads all 19 sensors at once.

There are only 121 kids but tons of little recording snippets, so the biggest decision was how to split them for testing. I split **by kid** — no kid shows up in both training and testing — and the model got **91.7% right on 24 kids it had never seen**, with a strong separating score of **0.965**. Unlike my ECG project, this score actually held up under honest testing, which was a relief.

2. Why I built this

EEG plus ADHD is a popular AI topic, and also a famous trap: a lot of projects mix snippets from the same kid into both training and testing, so the model just learns to recognize that specific kid instead of learning anything about ADHD. I wanted to avoid that.

To be super clear: **ADHD is diagnosed by doctors, never by a quick brain scan**. This just tells apart two groups in a research study — it's a pattern-matching demo, not a diagnosis.

3. The data

I used a public dataset (from Kaggle) with brain-wave recordings from 61 kids with ADHD and 60 without, ages 7 to 12, recorded while they did an attention task. I chopped each recording into 2-second snippets and — importantly — kept every kid entirely on one side of the split.

Group	Kids	Snippets
Training	83	5,838
Checking	14	948
Testing (new kids)	24	1,614

4. How it works

It's a neural network (193,000 numbers) that reads all 19 sensors together. And I graded it **by kid, not by snippet**: I average all of a kid's snippet-guesses into one final answer for that kid, because otherwise a kid with lots of snippets would count too much and mess up the score.

5. Results

On kids it never saw	Score
Right per kid	91.7% (22 of 24)
Separating score	0.965
Right per snippet	89.0%

The checking score and the real test score are close, which is a good sign the model learned something real about the two groups instead of just memorizing individual kids.

6. Getting it running

I put it in a web demo that draws all 19 brain-wave channels like a real hospital readout and sorts a recording on your device, plus an Android app. Both match the original model.

7. Honest limitations

- It tells apart two research groups — it does NOT diagnose any real person.
- Only 24 test kids, so 91.7% is really '22 out of 24'. Small numbers, so the true accuracy could be higher or lower.
- It's one study and one task; real attention issues are way more varied.

8. What's next

- Check which brain-wave frequencies carry the signal.
- Turn off different sensors to see which parts of the head matter most.
- Test on all 121 kids in rotation to get a more solid number.